



**Time:** 5 minutes

**Outcome:** Calculate Average Costs: AFC, AVC & ATC.

**Difficulty:** Intermediate

## Average Costs: AFC, AVC & ATC



### THE CORE IDEA

Average fixed cost (AFC) is fixed cost per unit; average variable cost (AVC) is variable cost per unit. Average total cost (ATC) is their sum:  $ATC = AFC + AVC$ . AFC falls as output spreads fixed cost over more units. Marginal cost (MC) pulls each average down when below it and up when above it.



### HOW TO RECOGNIZE IT

- ATC equals AFC plus AVC.
- The ATC-AVC gap is fixed cost per unit and shrinks as output increases.
- MC below an average pulls it down; MC above it pulls it up.

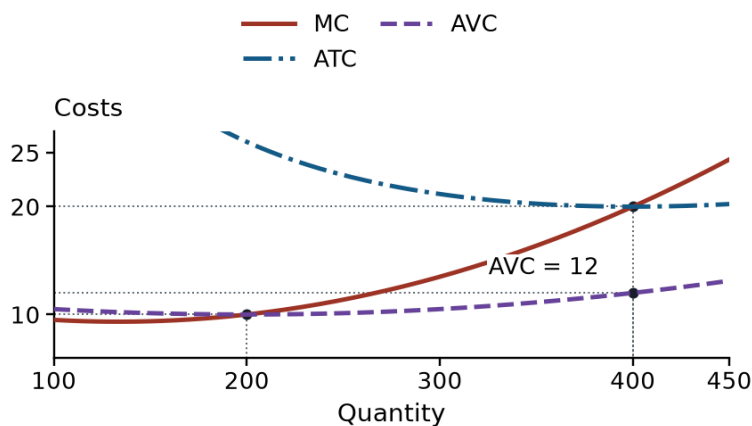


### WATCH OUT

AVC is the thick dashed curve; the fine dotted lines are reading guides. The vertical gap between ATC and AVC is AFC, not marginal cost.



### WORKED EXAMPLE: READING AVERAGE-COST CURVES



At 400 units, the graph shows average cost of \$20 and average variable cost of \$12. Because  $AC = AVC + AFC$ , average fixed cost is  $\$20 - \$12 = \$8$  per unit. The gap between AC and AVC is therefore the firm's average fixed cost.



### CHECK YOURSELF

At  $Q = 30$ , AVC is \$14 and ATC is \$24. Calculate AFC and total fixed cost. Verify your answer by multiplying the per-unit gap by output.

**READY?**

Return to the game and master this concept.